



Lyntra Data

India's #1 Training Institute

LLMOps Training

"1,00,000+ uplifted through our hybrid classroom & online training, enriched by real-time projects and job support."



3 Months



5 Live Projects



4.9/5

Lyntra Data Alumni Works @

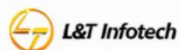


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1. Foundations of LLMs & LLMOps

Module 1: Introduction to Large Language Models

- Evolution of NLP and foundation models
- Transformer architecture fundamentals
- Tokens, embeddings, attention mechanisms
- Pretraining vs fine-tuning
- Generative AI use cases across industries

Module 2: What is LLMOps?

- Definition and scope of LLMOps
- LLMOps vs MLOps vs DevOps
- Challenges: scale, cost, latency, hallucinations
- Production LLM lifecycle
- Organizational readiness for LLM adoption

Module 3: LLM System Architecture

- End-to-end LLM application architecture
 - Model layer, orchestration layer, application layer
 - API-based vs self-hosted models
 - Stateless vs stateful LLM systems
 - Multi-model and fallback architectures
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2. Programming & Frameworks for LLM Applications

Module 1: Python for LLM Engineering

- Python refresher for AI engineers
- Async programming for LLM calls
- Token counting and rate limiting
- Error handling and retries

Module 2: LLM Frameworks

- LangChain architecture and components
- LlamaIndex for data-aware LLM apps
- Prompt templates and chains
- Agents and tool calling

Module 3: API-Based LLM Integration

- Hosted LLM APIs (OpenAI, Anthropic, Google)
- Authentication and key management
- Streaming responses
- Cost and latency optimization

Module 4: Building LLM Applications

- Chatbots and copilots
 - Retrieval-Augmented Generation (RAG)
 - Document Q&A systems
 - Function calling and tool integration
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3. Data Management & Prompt Engineering

Module 1: Prompt Engineering Fundamentals

- Prompt structure and components
- Zero-shot, few-shot prompting
- Chain-of-thought prompting
- Prompt versioning and testing

Module 2: Advanced Prompt Techniques

- Role prompting and system prompts
- Self-consistency and reflection
- Prompt compression
- Guardrails and constraints

Module 3: Data for LLM Applications

- Structured vs unstructured data
- Chunking and indexing strategies
- Embeddings and vector databases
- Metadata and filtering

Module 4: Vector Databases & Retrieval

- FAISS, Pinecone, Weaviate, Chroma

- Similarity search techniques
 - Hybrid search (keyword + vector)
 - RAG performance optimization
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4. LLM Training, Fine-Tuning & Adaptation

Module 1: Fine-Tuning Fundamentals

- When to fine-tune vs prompt engineer
- Supervised fine-tuning (SFT)
- Dataset preparation and labeling
- Evaluation before and after fine-tuning

Module 2: Parameter-Efficient Fine-Tuning (PEFT)

- LoRA and adapters
- QLoRA and memory-efficient tuning
- Training at scale
- Cost vs performance trade-offs

Module 3: Reinforcement Learning from Human Feedback (RLHF)

- Preference datasets
- Reward models
- Policy optimization overview
- Safety alignment concepts

Module 4: Open-Source LLMs

- LLaMA, Mistral, Falcon overview
- Model licensing considerations
- Model selection criteria

- Hosting open-source LLMs
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5. LLM Deployment & Inference Engineering

Module 1: Model Serving Architectures

- Batch vs real-time inference
- REST vs gRPC serving
- Stateless inference services
- Multi-region deployment

Module 2: Performance Optimization

- Quantization techniques
- Model sharding and parallelism
- GPU vs CPU inference
- Caching strategies

Module 3: Containerization & Orchestration

- Dockerizing LLM services
- Kubernetes for LLM workloads
- Autoscaling strategies
- GPU scheduling and resource isolation

Module 4: Inference Frameworks

- vLLM, TGI, Triton Inference Server
 - Ray Serve
 - Model routing and fallback
 - Canary and blue-green deployments
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6. LLMOps: Monitoring, Evaluation & Governance

Module 1: LLM Observability

- Latency, throughput, token usage
- Cost monitoring
- User behavior analytics
- System health dashboards

Module 2: Evaluation & Testing

- Offline evaluation metrics
- Online A/B testing
- Prompt and model regression testing
- Golden datasets

Module 3: Safety, Security & Governance

- Hallucination detection
- Content moderation
- Prompt injection attacks
- PII handling and compliance

Module 4: Feedback Loops & Continuous Improvement

- Human-in-the-loop systems
 - Feedback collection strategies
 - Active learning
 - Automated retraining pipelines
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7. Advanced LLMOps & Enterprise Systems

Module 1: Multi-Agent Systems

- Agent orchestration patterns
- Planning and memory
- Tool-using agents
- Failure handling in agents

Module 2: Enterprise LLM Platforms

- Internal LLM gateways
- Model catalogs and registries
- Access control and RBAC
- Cost allocation and chargeback

Module 3: Hybrid & On-Prem LLMOps

- Private cloud deployments
- Edge inference
- Data residency requirements
- Hybrid inference architectures

Module 4: LLM Cost Optimization

- Token budgeting
 - Dynamic model selection
 - Prompt optimization
 - ROI measurement
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8. Capstone Project

End-to-End Production LLM System - Business use case definition - RAG-based LLM application - Prompt and model versioning - Monitoring, evaluation, and safety controls - Scalable deployment on Kubernetes - Cost, latency, and quality optimization